

TOUGGOURT, Algeria

WMO: 605550

Lat: 33.068N

Lon: 6.089E

Elev: 85

StdP: 100.31

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.4	3.8	-5.8	2.3	16.9	-4.1	2.7	14.5	12.4	13.8	10.6	14.6	1.0	180	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	45.1	22.5	43.7	22.2	41.9	22.0	25.4	37.3	24.6	36.8	23.8	36.3	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.1	31.0	21.0	15.9	30.6	20.0	14.8	30.7	78.1	37.0	74.5	36.4	71.7	36.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.8	9.2	7.8	DB	-0.1	47.6	1.1	1.0	-0.9	48.3	-1.5	48.9	-2.1	49.5	-2.9	50.3	
			WB	-1.3	27.5	1.1	1.1	-2.1	28.3	-2.7	28.9	-3.3	29.6	-4.1	30.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	11.2	13.0	17.1	21.3	26.4	31.2	34.4	33.7	29.0	23.4	16.4	12.1
	DBStd	8.73	2.28	2.95	3.46	3.65	3.73	3.42	2.67	2.51	3.51	3.40	3.06	2.46
	HDD10.0	27	13	5	0	0	0	0	0	0	0	0	1	8
	HDD18.3	718	221	151	64	12	0	0	0	0	2	74	195	
	CDD10.0	4591	51	90	222	339	509	637	758	735	571	416	193	72
	CDD18.3	2241	0	3	27	100	251	387	499	477	321	159	16	1
	CDH23.3	33162	2	39	331	1144	3268	5963	8433	7836	4373	1624	143	6
	CDH26.7	21015	0	7	97	487	1810	3896	6009	5422	2554	706	27	1
Wind	WSAvg	3.1	2.7	3.0	3.5	3.9	3.9	3.4	3.1	3.0	2.9	2.6	2.5	2.4
Precipitation	PrecAvg	69	21	6	6	6	2	2	1	3	9	4	4	5
	PrecMax	169	101	45	26	38	26	9	13	19	34	24	31	25
	PrecMin	26	5	2	0	0	0	1	0	1	3	0	0	0
	PrecStd	40	25	9	7	9	6	2	2	4	8	7	7	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	27.1	32.8	37.6	42.2	45.6	47.3	46.2	42.8	37.7	29.7	23.4
		MCWB	13.1	15.1	15.8	18.0	20.9	22.1	22.8	23.0	21.9	19.4	16.7	13.5
	2%	DB	20.1	24.1	29.1	34.3	39.2	43.7	45.8	44.6	40.2	35.0	26.8	21.0
		MCWB	12.0	13.3	15.3	17.4	20.2	21.7	22.7	22.6	21.6	19.5	15.9	12.9
	5%	DB	18.5	21.8	26.8	31.9	36.8	41.5	44.1	42.9	38.2	32.6	24.8	19.4
		MCWB	11.3	12.6	14.3	16.7	19.6	21.4	22.5	22.2	21.6	19.0	15.2	12.3
	10%	DB	17.1	19.8	24.8	29.2	34.8	39.2	42.1	41.0	36.2	30.5	22.9	18.0
		MCWB	10.7	11.6	13.8	16.2	19.3	21.0	22.2	22.5	21.4	18.8	14.5	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	16.3	18.2	20.4	23.7	25.4	26.8	26.6	25.3	23.4	18.6	15.2
		MCDB	20.6	24.6	25.3	30.7	34.5	37.1	38.4	38.0	34.6	29.5	26.1	20.3
	2%	WB	13.0	14.4	16.4	18.8	22.1	23.9	25.2	25.6	24.2	21.7	17.5	14.1
		MCDB	18.4	22.3	26.3	29.4	34.7	37.3	38.3	37.4	33.7	29.5	24.9	19.0
	5%	WB	12.0	13.1	15.4	17.9	21.0	23.0	24.3	24.7	23.4	20.7	16.2	13.1
		MCDB	17.2	20.1	24.7	28.5	33.3	36.5	38.2	36.7	33.4	29.1	22.9	18.2
	10%	WB	11.1	12.1	14.4	17.0	20.1	22.1	23.5	23.9	22.6	19.7	15.1	12.1
		MCDB	16.1	18.7	23.2	27.3	32.3	36.1	38.0	36.1	32.7	28.0	21.5	17.3
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.9	13.4	13.4	13.5	13.6	14.1	13.5	12.5	12.7	12.4	12.1
		MCDBR	13.9	15.4	16.0	16.5	15.9	15.8	16.3	15.8	14.6	15.0	13.8	13.4
		MCWBR	7.8	7.6	7.1	6.8	6.2	5.6	5.7	5.4	5.3	6.0	6.4	7.4
	5% WB	MCDBR	12.7	13.6	14.1	14.3	13.9	13.8	13.9	13.3	12.5	12.3	12.9	12.1
	MCWBR	7.7	7.5	7.0	6.6	6.5	6.2	6.2	6.1	5.9	5.9	6.8	7.4	
Clear-Sky Solar Irradiance	taub	0.362	0.393	0.498	0.546	0.604	0.608	0.644	0.610	0.589	0.512	0.419	0.378	
	taud	2.122	2.021	1.745	1.665	1.576	1.578	1.532	1.597	1.650	1.787	1.972	2.068	
	Ebn at Noon	828	841	775	759	721	715	687	701	689	707	749	779	
	Edh at Noon	124	152	218	247	273	271	283	261	237	191	144	125	
All-Sky Solar Radiation	RadAvg	3.51	4.47	5.55	6.64	7.19	7.64	7.56	6.82	5.72	4.60	3.62	3.11	
	RadStd	0.15	0.18	0.23	0.26	0.23	0.20	0.12	0.16	0.16	0.20	0.12	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	-15	N/A	+132	N/A

Nomenclature: See separate page